

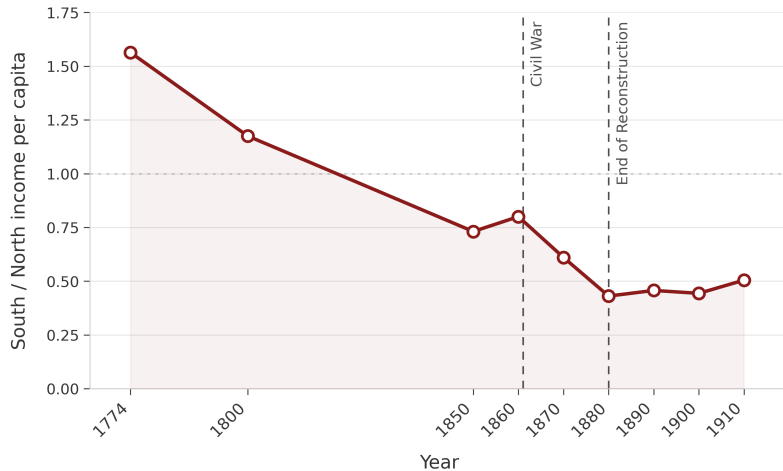
American Slavery, Innovation, and Economic Growth

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Intro

Reversal of Fortunes in the United States: Decline of the South



Data: Lindert & Williamson (2012) and Klein (2013).

This paper asks if slavery can explain the differential growth rate between the North and the South throughout the 19th century.

Technological change is the main driver of economic growth

Labor coercion in the South under Slavery

- Reduces relative input unit cost of labor
- Weakens incentives to innovate
- **Lowers the rate of productivity growth**

The preview of paper and findings

1. **Empirics:** low growth, less patenting and mechanization before emancipation
 - Upon emancipation, increased patenting in tasks formerly performed by enslaved workers
2. **Model:** Endogenous growth model with tasks, labor coercion, and automation.
 - Coerced labor unit costs affect innovation incentives and long-term growth
 - Model generates all the empirical results
3. **Quantification:**
 - Calibrated model to two regions: North and South
 - Calibrated model delivers the observed pattern of regional income differences
 - Counterfactual results: huge economic gains for the South from earlier abolition

Contributions

1. Directed technological change

Habakkuk (1962); Acemoglu (1998, 2002, 2010); Hornbeck and Naidu (2014); Hanlon (2015)

Contribution: Slavery distorts incentives to develop labor-saving technologies.

2. Slavery and Growth

Fogel and Engerman (1974); Ransom and Sutch (1977); Olmstead and Rhode (2008); Acemoglu-Wolitzky (2011); Feigenbaum et al.(2022) Allen et al. (2025)

Contribution: Once technological change is endogenous slavery could slow growth.

3. Growth Theory

Acemoglu and Autor (2011); Acemoglu and Restrepo (2018a,b); Jones and Liu (2024); Jones and Tonetti (2026)

Contribution: A GE growth model in which relative unit costs have permanent growth effect

Evidence

Fact 2: Lower capital share in more slave populated states

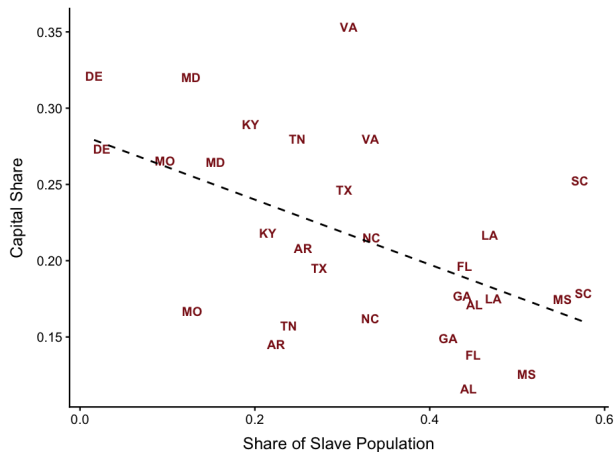
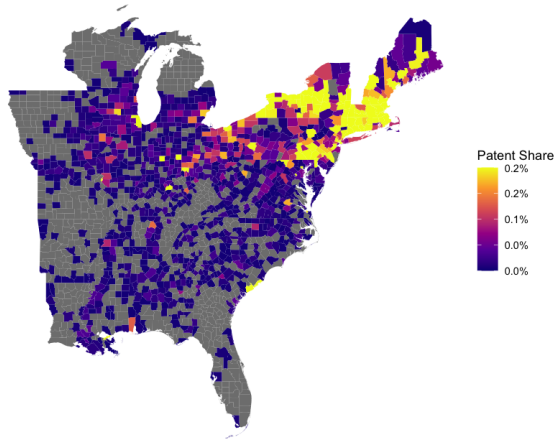


Figure 1: Data: US Census of Agriculture

Fact 3a: Less patenting in the South



Data: Petrala et al(2016)

A task is a unit of productive activity.

Examples

- Agricultural Crop: Digging → planting → cultivation → harvesting → ginning/processing

Innovations in Agricultural Implements

X82-77

UNITED STATES PATENT OFFICE.

CYRUS H. McCORMICK, OF ROCKBRIDGE COUNTY, VIRGINIA.

IMPROVEMENT IN MACHINES FOR REAPING SMALL GRAIN.

Specification forming part of Letters Patent dated June 21, 1834

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Cyrus McCormick Patent

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Letters Patent No. 68,369, dated April 2, 1857.

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John Deere Steel Plow Patent



Eli Whitney cotton gin (1793)



McCormick mechanical reaper (1831)



John Deere steel plow (1837)

Fact 3b: Low patenting in slave performed tasks

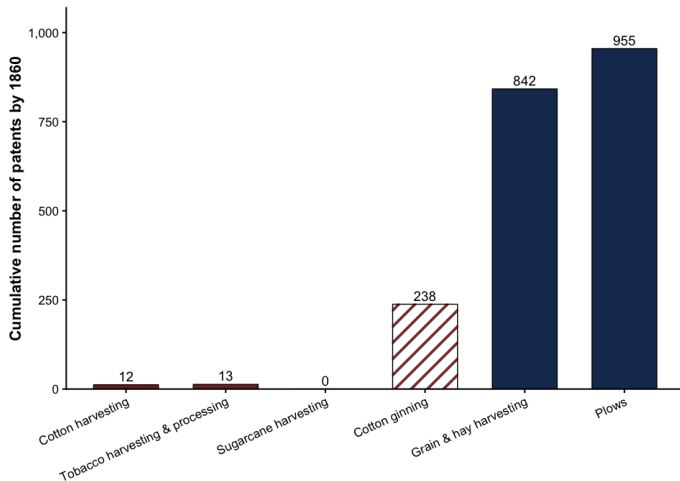


Figure 2: Number of patents by tech class; Data: USPTO

Emancipation: the exogenous event that removes coercion

- **Patent Data:** USPTO
- **Treated** (61 subclasses): Cotton, sugarcane, tobacco harvesting & processing
- **Control** (1,382 subclasses): Grain/hay harvesting, plows, threshing, drills classification
- 5-year bins, 1845–1910 | 14 periods | 20,202 cells | 41,522 patents patents

Poisson QMLE Event Study:

$$\mathbb{E}[Patent_{ct}] = \exp\left(\alpha_c + \lambda_t + \sum_{k \neq -5} \beta_k \cdot 1[k = t] \cdot Treated_c\right)$$

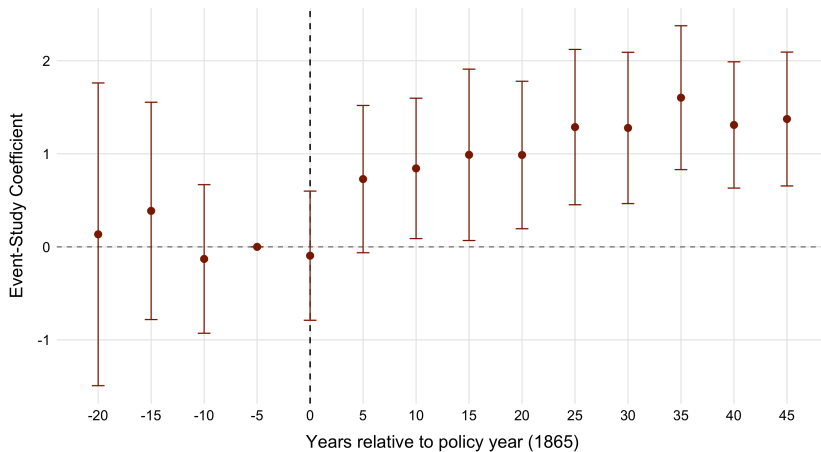
α_c Subclass FE

λ_t Period FE

$$\log \text{incidence rate ratio} : \beta_k = \log \frac{\mathbb{E}[N_{ct} | \mathbf{Tr}=1]}{\mathbb{E}[N_{ct} | \mathbf{Tr}=0]} - \log \frac{\mathbb{E}[N_{c,-5} | \mathbf{Tr}=1]}{\mathbb{E}[N_{c,-5} | \mathbf{Tr}=0]}$$

◀ exogeneity

Emancipation increases patenting in former slave tasks



patents

placebo tests

robustness

robustness bins

robustness ref

Model

Tasks and Production

- Final output combines a unit continuum of tasks, $i \in [0, 1]$.
- Each task can be performed by capital, low-skill labor, or high-skill labor.

Labor Markets

- Labor markets are perfectly competitive.
- Under slave institutions, low-skill labor can be coerced.
- Coercion alters the relative unit price of labor to capital.

Innovation

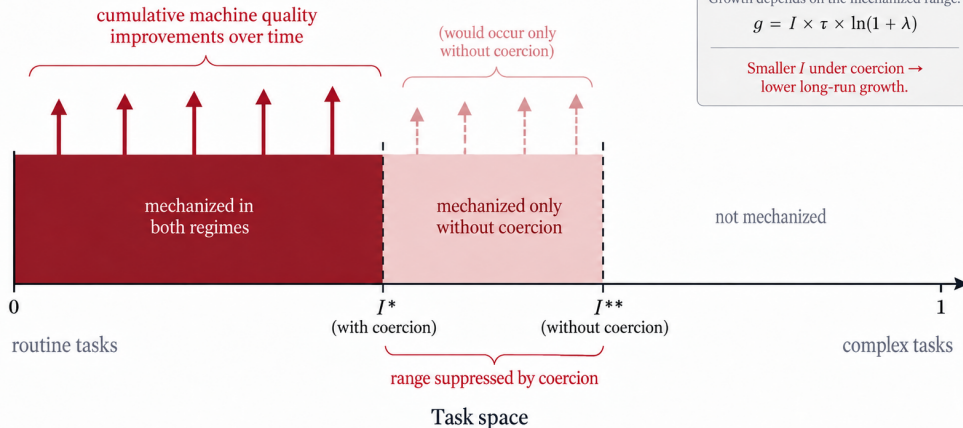
- Task producers operating with capital innovate to develop the best machine and become monopolists.

Households

- The representative household supplies fixed high-skill (H) and low-skill (L) labor.

Model Summary: Task Space, Automation, and Effect of Coercion

Coercion suppresses the range of tasks that benefit from cumulative machine productivity growth.



Coercion Technology and labor Productivity

Skill productivity: high-skill workers have an absolute productivity advantage,

$$A_H > A_L.$$

Human labor productivity at task i :

$$\gamma(i), \quad \gamma'(i) > 0$$

Coercion technology: coercion applies to low-skill labor and augments effective labor input,

$$c(i) = \begin{cases} 1, & \text{without coercion,} \\ \geq 1, & \text{under coercion,} \end{cases} \quad c'(i) < 0, \quad \frac{d}{di} [c(i)\gamma(i)] > 0.$$

Effective labor productivity in task i

$$\text{Low skill: } A_L c(i) \gamma(i)$$

$$\text{High skill: } A_H \gamma(i)$$

Coercion Technology and labor Productivity

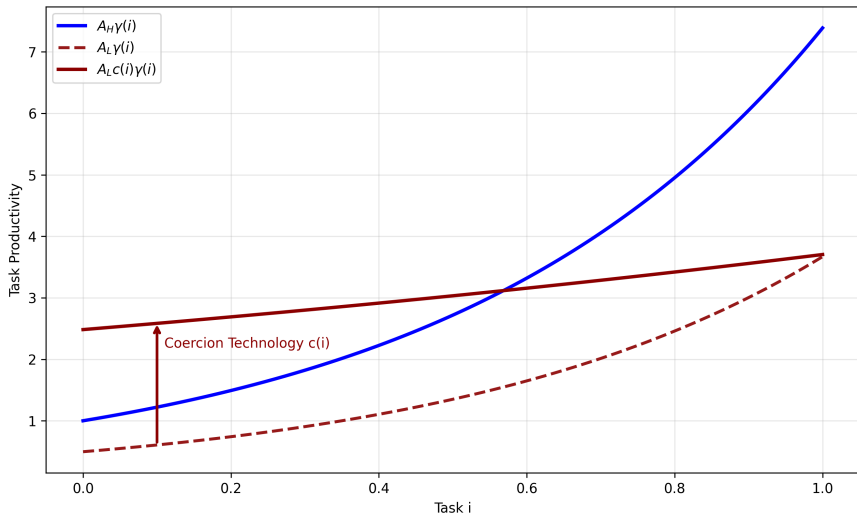


Figure 3: Coercion Technology and Productivity Functions

Final output combines a unit continuum of tasks, $i \in [0, 1]$:

$$\ln Y_t = \int_0^1 \ln y_t(i) di.$$

Task Producers:

$$y_t(i) = \begin{cases} A_L \mathbf{c}(i) \gamma(i) l_t(i) + A_H \gamma(i) h_t(i), & \text{for all } i \in [0, 1]; \\ A_L \mathbf{c}(i) \gamma(i) l_t(i) + A_H \gamma(i) h_t(i) + \mathbf{q}_t(i) \mathbf{k}_t(i), & \text{for all } i \in [0, I_t], \end{cases} \quad (1)$$

Capital Input producer:

- $q'_0(i) < 0$
- $q_{t+\Delta t}(i) = (1 + \lambda)q_t$

Task Allocation

Firms assign each task to the input with the lowest unit cost:

$$\chi_K(i) = \frac{\psi w_H}{q_0(i)}, \quad \chi_L(i) = \frac{w_L}{A_L c(i) \gamma(i)}, \quad \chi_H(i) = \frac{w_H}{A_H \gamma(i)}.$$

Monotonic productivity schedules imply two cutoffs, $I^* < \bar{I}$:

$$\frac{\psi w_H}{q_0(I^*)} = \frac{w_L}{A_L c(I^*) \gamma(I^*)} \quad \text{and} \quad \frac{w_L}{A_L c(\bar{I}) \gamma(\bar{I})} = \frac{w_H}{A_H \gamma(\bar{I})}.$$



Productivity curves

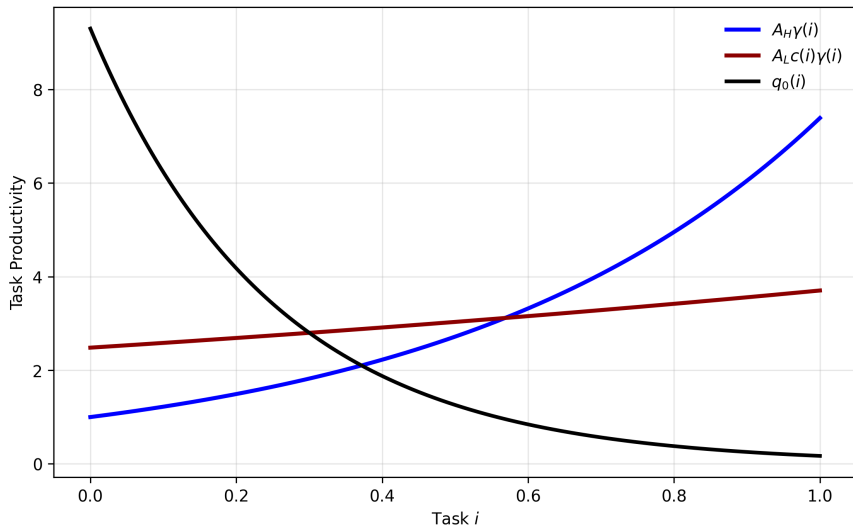


Figure 4

- **Task Prices**

- Labor intensive task prices = marginal cost
- Capital intensive tasks $1/\phi$ * marginal cost
- Using final good demand $\implies$ Optimal input demand
- Labor market clearing implies

$$w_{L,t} = [\bar{I} - I^*] \frac{Y_t}{L}, \quad w_{H,t} = [1 - \bar{I} + \phi I^*] \frac{Y_t}{H}.$$

derivations

Stationary Equilibrium

The equilibrium is characterized by the two cutoff conditions:

$$\frac{\bar{I} - I^*}{1 - \bar{I} + \phi I^*} \frac{H}{L} q(I^*) = \psi A_L \gamma(I^*) c(I^*), \quad (1)$$

$$\frac{\bar{I} - I^*}{1 - \bar{I} + \phi I^*} \frac{H}{L} A_H = A_L c(\bar{I}). \quad (2)$$

$\implies$ **coercion-adjusted automation threshold (CAAT)**

$$\frac{q_0(I^*)}{\gamma(I^*)} = \psi A_H \underbrace{\frac{c(I^*)}{c(\bar{I})}}_{>1 \text{ under coercion}}. \quad (\text{CAAT})$$

LHS decreasing in $I \implies$ coercion decreases mechanization I^*

Equilibrium Productivity distributions

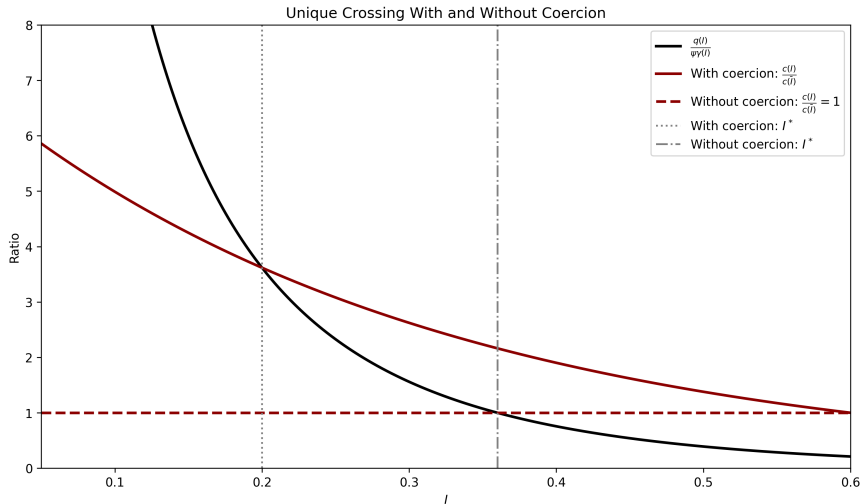


Figure 5

Vertical Innovation

$$\underbrace{r_t V_t(i)}_{\text{return}} - \dot{V}_t(i) = \underbrace{\pi_t(i)}_{\text{profit}} - \underbrace{\tau_t(i) V_t(i)}_{\text{creative destruction}}$$

Entrant: $\max_x \left\{ \underbrace{x_t(i) V_t(i)}_{\text{expected value}} - \underbrace{\kappa x_t(i)^\alpha Y_t}_{\text{R\&D cost}} \right\}$

BGP
 $\implies$

$$\boxed{\tau^*}$$

Vertical Innovation

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$\xRightarrow{\text{BGP}}$

$$\tau^*$$

τ^* : innovation within mechanized tasks

Mechanization

$$\underbrace{r_t V_t^m(I_t)}_{\text{return}} - \dot{V}_t^m(I_t) = \underbrace{\pi_t^m(I_t)}_{\text{profit}} - \underbrace{\tau_t V_t^m(I_t)}_{\text{creative destruction}}$$

Entrant: $\max_{x^m} \left\{ \underbrace{x_t^m(I_t) V_t^m(I_t)}_{\text{expected value}} - \underbrace{\kappa^h [x_t^m(I_t)]^\alpha Y_t}_{\text{R\&D cost}} \right\}$

$$I_t \rightarrow I^* \Rightarrow \pi_t^m(I_t) \rightarrow 0 \Rightarrow x_t^m \rightarrow 0$$

I^* : long-run mechanization frontier

Long-Run Growth

Aggregate output. Summing across equilibrium task outputs,

$$Y_t = Q_t \left(\frac{L}{\bar{I} - I^*} \right)^{\bar{I} - I^*} \left(\frac{H}{1 - \bar{I}} \right)^{1 - (\bar{I} - I^*)},$$

where

$$Q_t \equiv \exp \left[\int_0^{I^*} \ln q_t(i) di + \int_{I^*}^{\bar{I}} \ln (A_L c(i) \gamma(i)) di + \int_{\bar{I}}^1 \ln (A_H \gamma(i)) di \right] \left(\frac{\phi}{\psi} \right)^{I^*}.$$

Balanced-Growth Rate

$$g^* = \underbrace{I^*}_{\text{mechanized tasks}} \underbrace{\tau^*}_{\text{vertical innovation rate}} \underbrace{\ln(1 + \lambda)}_{\text{step size}}$$

Central Theoretical Results

1. Coercion reduces the long-run mechanization frontier.

Fact 2

$$I^* < I^{**}$$

2. A smaller mechanized domain permanently lowers long-run growth.

Fact 1

$$g^* = I^* \tau^* \ln(1 + \lambda) < I^{**} \tau^* \ln(1 + \lambda) = g^{**}$$

3. Coercion reduces long-run expenditure on machine innovation.

Fact 3

$$\frac{R_t^*}{Y_t} = I^* \kappa(\tau^*)^\alpha < I^{**} \kappa(\tau^*)^\alpha = \frac{R_t^{**}}{Y_t}$$

4. Removing coercion creates new profitable mechanization opportunities.

Fact 4

Increasing Innovation in the the tasks $I^{**} - I^*$

A quantitative exercise: the US North and South

- **Regional goods.** Y^f for $f \in \{N, S\}$, aggregated into a final good

$$Y_t = \left[\chi^{\frac{1}{\varepsilon}} (Y_t^N)^{\frac{\varepsilon-1}{\varepsilon}} + (1-\chi)^{\frac{1}{\varepsilon}} (Y_t^S)^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}}, \quad \chi \in (0, 1), \varepsilon > 0.$$

- **Segmented markets.** Households
 - Households consume the final good but save in regional assets: $r^S \neq r^N$.
 - No labour mobility: $w_\ell^S \neq w_\ell^N$.
- **Key asymmetry.** Coercion in the south $c(i) > 1$, and no coercion in the north $c(i) = 1$

Calibration: matched moments

The moment-discrepancy vector is

$$g(\Theta) = \begin{bmatrix} \text{capitalshare}^{1800}(\Theta) - 0.1448 \\ \text{capitalshare}^{1860}(\Theta) - 0.2511 \\ \text{capitalshare}^{1890}(\Theta) - 0.2834 \\ \text{capitalshare}^{1900}(\Theta) - 0.2956 \\ \Delta \ln(y_S / y_N)_{1800-1860}(\Theta) - (-0.3473) \\ \ln(w_H / w_L)_{1865}(\Theta) - \ln(1.69) \\ \ln(Q^C / Q^F)_{1865}(\Theta) - \ln(1.35) \end{bmatrix}.$$

The parameter vector is chosen to minimize

$$\hat{\Theta} = \arg \min_{\Theta} g(\Theta)' W g(\Theta)$$

Convergence to the new steady state, $I^* \rightarrow I^{**}$

Calibration: externally set

Parameter / Target	Symbol	Value	Source / Role
Discount rate	ρ	0.020	Standard
R&D curvature	α	2	Akcigit & Kerr (2018)
Profitability	ϕ	0.900	Vedder and Gallaway (1980)
Innovation step	λ	1/9	Consistent with $\phi = (1 + \lambda)^{-1}$
High-skill productivity	A_H	1.000	Normalization
Low-skill productivity	A_L	0.43	Margo & Goldin(1992)
Prewar labor composition	L/H	1.040	US census 1860
Postwar labor composition	L^+ / H	0.707	Ransom & Sutch (2001)
CES elasticity	ε	5.000	Benchmark elasticity
Long-run Northern growth	g_N^*	1.23%	Pins down τ^* conditional on I^{**}

Calibration: estimated and implied objects

Object	Symbol	Value	Identification / Construction
<i>Jointly estimated</i>			
Free mechanization frontier	I^{**}	0.3284	Capital-share levels and transition
Coercive/free ratio	I^* / I^{**}	0.5036	Prewar North–South growth gap
Coercive mechanization	I^*	0.1654	$I^* = (I^* / I^{**}) I^{**}$
Prewar labor cutoff	$\bar{I}$	0.5400	Hire-cost ratio $w_H / w_L = 1.69$
Horizontal/vertical R&D cost	ζ	82.59	Speed of mechanization transition
Initial Northern position	f_N	0.4888	$I_N(1800) = f_N I^{**}$
Initial Southern position	f_S	0.9748	$I_S(1800) = f_S I^*$
Coercion-wedge slope	β_c	3.4227	$Q^C / Q^F = 1.35$
<i>Implied</i>			
Coercion-wedge level	C	8.737	Cutoff conditions
Combined task slope	$\beta + \beta_q$	3.657	Jointly implied by I^* and I^{**}
BGP innovation intensity	τ^*	0.3555	$g_N^* = I^{**} \tau^* \ln(1 + \lambda)$
Vertical R&D cost	κ	0.3746	Terminal HJB condition
Horizontal R&D cost	κ_I	30.94	$\kappa_I = \zeta \kappa$

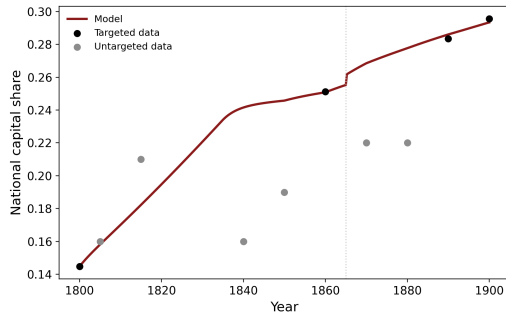


Figure 6: Capital Share

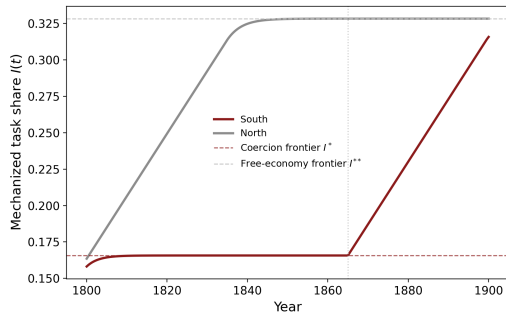


Figure 7: Mechanization

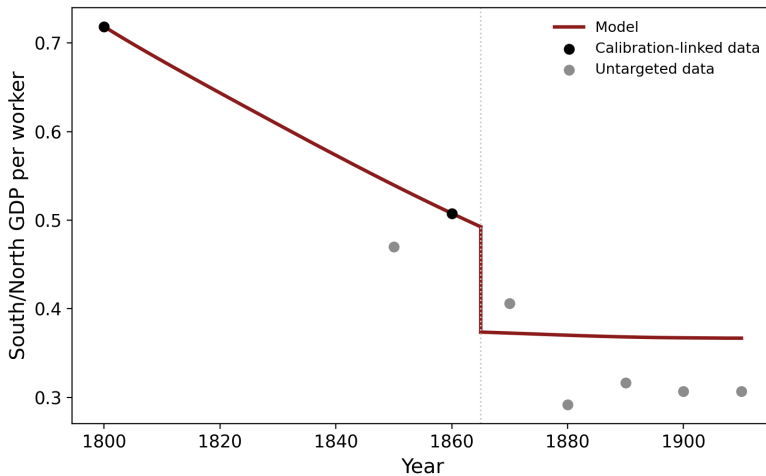
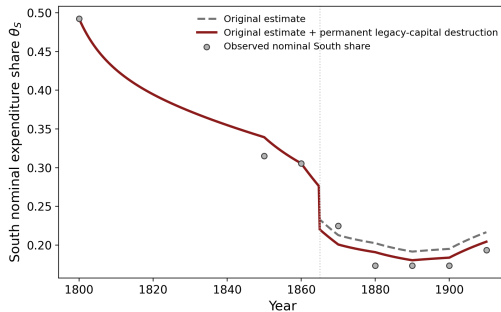
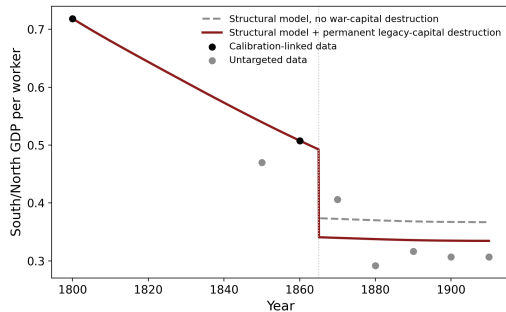
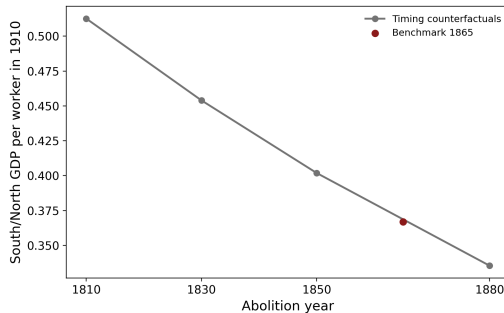
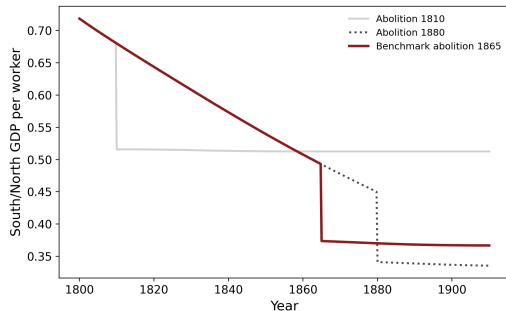


Figure 8

Model fit With Capital Destruction



Counterfactual: Early Abolition brings huge economic gains to the South



Conclusion

- American slavery suppressed incentives for labor-saving innovation.
 - 3 times more patenting in slave-intensive tasks upon emancipation
- Slavery imposes substantial costs on long-run economic growth.
 - 38 % higher GDP per worker with an early abolition in 1810
- The framework can be applied more broadly to study how labor costs affect innovation
 - Unions, monopsony power

Appendix

Appendix: Historical Background



Appendix: Historical Background



Appendix: Enslaved Labor

- **30-40%** of the total southern population. $\approx$ **12-18%** of total population
- Agriculture $\approx$ **87%** (41% Cotton) of total Enslaved Production in 1860
- **98%** of cotton production in the southern slave plantations.
- Cotton **57%** of total US exports and $\approx$ **5%** of GNP in 1860.

Appendix: Historical Background

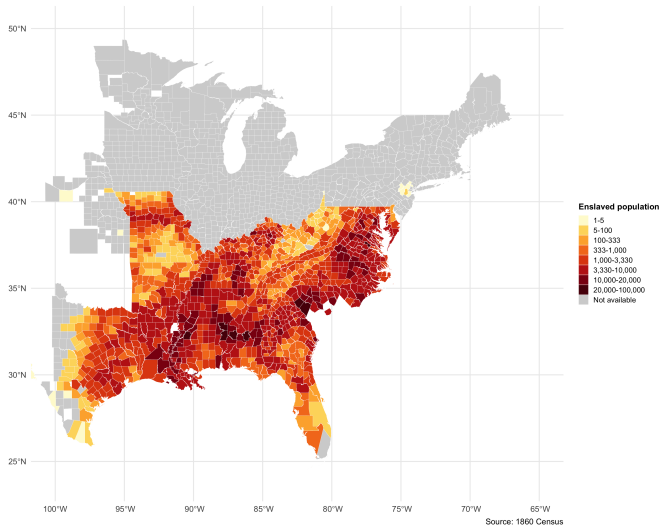
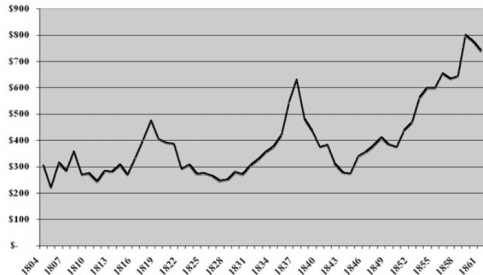
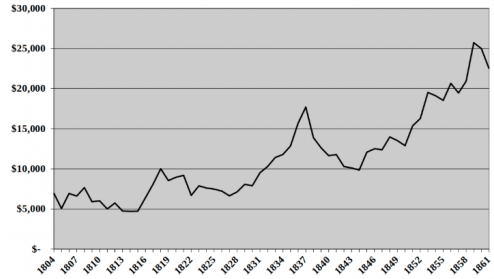


Figure 9: Slave Population Distribution in the US in 1860.

Appendix: Slave Prices



Average Price of a Slave Over Time Current dollars



The Real Price of Owning a Slave in 2020 Dollars

Figure 10: (Williamson & Cain, 2026)

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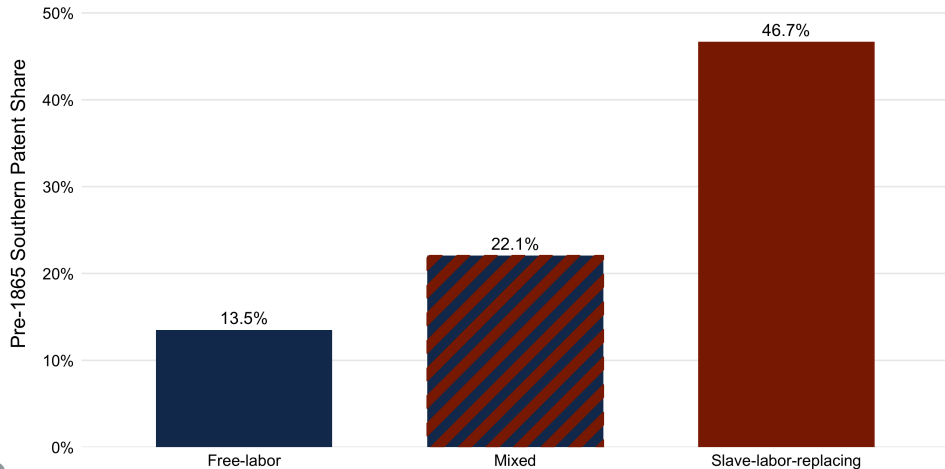
Appendix: Patent Classification: Four Technology Groups

Patents are assigned to four groups using **USPC** codes based on their relationship to slave vs. free labor:

Group	Label	USPC Codes
G1	Slave Replacing	Cl. 56 (sub. 27.5, 28–50, 62); Cl. 131 ($\geq 280 \leq 327$). Cotton picking, tobacco & sugarcane harvest, tobacco processing.
G2	Mixed Replacing	Cl. 56 (51–121, excl. 62); Corn Harvesting.
G3	Slave Complementary	Cl. 19 (1–67). Cotton ginning .
G4	Free-Labor Replacing	Cl. 56 ($<27.5, >121$); Cl. 172, 460, 111. Grain/hay harvest, plows, threshing, drills.

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Slave-Replacing Technologies Were More Likely to Originate in the South



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Geographic Validation Measures

1. Patent-weighted Southern share

$$\frac{\text{Pre-1865 patents from the South in group } g}{\text{Total pre-1865 patents in group } g}$$

2. Mean subclass Southern share

$$\frac{1}{J_g} \sum_{j \in g} \left(\frac{\text{Southern patents in subclass } j}{\text{Total patents in subclass } j} \right)$$

3. Mean subclass Southern share, $N_j \geq 2$

$$\frac{1}{J_g^{(2)}} \sum_{\substack{j \in g \\ N_j \geq 2}} \left(\frac{\text{Southern patents in subclass } j}{\text{Total patents in subclass } j} \right)$$

Same measure, excluding subclasses with only one observed patent.

Prior:

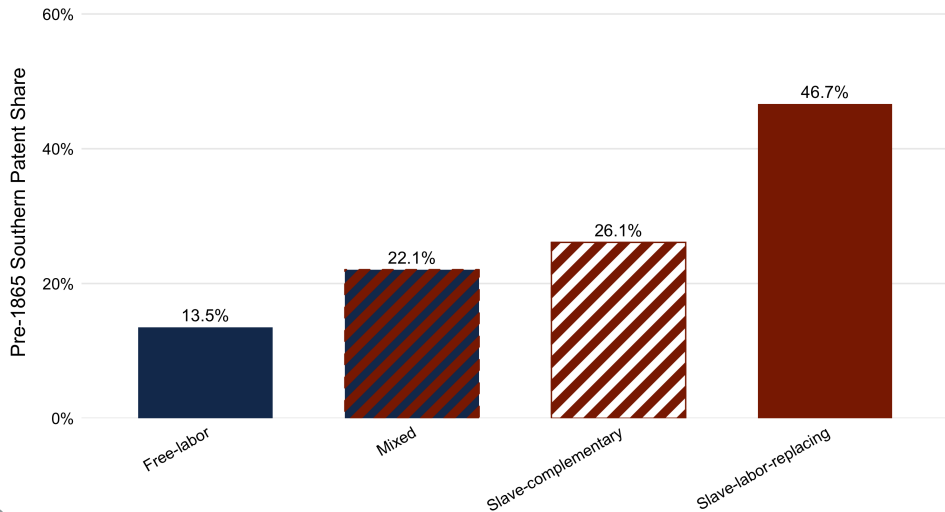
Slave-replacing > Mixed > Free-labor

Slave-Replacing Technologies Were More Likely to Originate in the South

Table 1: Pre-Emancipation Geographic Orientation of Technology Groups

Measure	Slave-replacing	Mixed	Free-labor
Patent-weighted South share	46.7%	22.1%	13.5%
Mean subclass South share	51.0%	16.0%	13.8%
Mean subclass South share, ≥ 2 patents	38.1%	29.6%	14.5%

Slave-Replacing Technologies Were More Likely to Originate in the South



Geographic Validation: Linear Probability Model

$$South_{ijt} = \alpha_t + \beta_1 SlaveReplacing_j + \beta_2 Mixed_j + \varepsilon_{ijt}$$

- $South_{ijt} = 1$ if patent i in subclass j and year t originates in the South; 0 if it originates in the North.
- Free-labor technologies are the reference category.

Geographic Validation: LPM Estimates

Dependent Variable:	South	
Model:	(1) Free omitted (1)	(2) Mixed omitted (2)
<i>Variables</i>		
Slave-replacing	0.3204*** (0.1204)	0.2384* (0.1270)
Mixed	0.0820* (0.0427)	
Free-labor		-0.0820* (0.0427)
<i>Fixed-effects</i>		
year_f	Yes	Yes
<i>Fit statistics</i>		
Observations	3,661	3,661
<i>Clustered (Subclasses) standard-errors in parentheses</i>		
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>		

Column (1) uses Free-labor technologies as the reference category. Column (2) uses Mixed technologies as the reference category.

Table 2: Log Cumulative Patents by Technology Group, 1860 vs. 1900

	1860	1900	$\Delta_{1900-1860}$
<i>log cumulative patents</i>			
Slave-Labor Replacing	4.143	6.733	2.590
Free-Labor Replacing	8.976	10.525	1.549

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Table 3: Log Cumulative Patents by Technology Group, 1860 vs. 1900

	1860	1900	$\Delta_{1900-1860}$
<i>log cumulative patents</i>			
Slave-Labor Replacing	4.143	6.733	2.590
Free-Labor Replacing	8.976	10.525	1.549
Total agriculture	9.007	10.711	1.704

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Table 4: Log Cumulative Patents by Technology Group, 1860 vs. 1900

	1860	1900	$\Delta_{1900-1860}$	$\Delta - \Delta^{\text{tot}}$
<i>log cumulative patents</i>				
Slave-Labor Replacing	4.143	6.733	2.590	+1.021
Mixed Labor Replacing	4.990	7.071	2.080	+0.511
Slave Complementary	5.994	7.412	1.418	-0.151
Free-Labor Replacing	8.976	10.525	1.549	-0.020

back

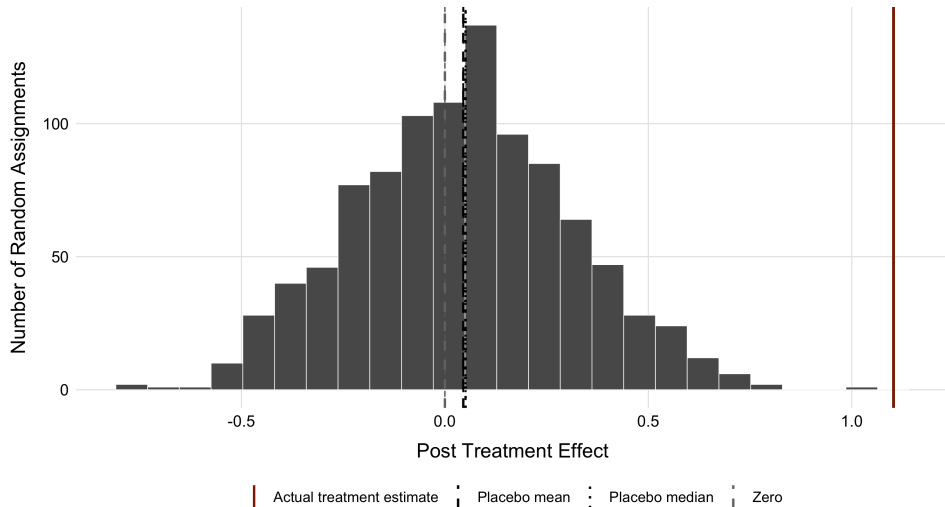
Change in Patenting in Harvesting Technologies by Labor type

Table 5: Log Cumulative Patents by Technology Group, 1860 vs. 1900

	1860	1900	$\Delta_{1900-1860}$	$\Delta - \Delta^{\text{tot}}$
<i>log cumulative patents</i>				
Cotton harvesting	3.258	5.820	2.562	+0.858
Cotton gin	5.994	7.412	1.418	-0.287
Other northern harvesting	7.798	9.203	1.406	-0.298
Total agriculture	9.007	10.711	1.704	—

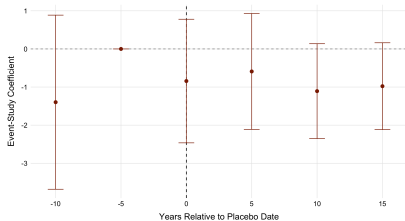
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Placebo Test : 1000 random draws of treated subclasses

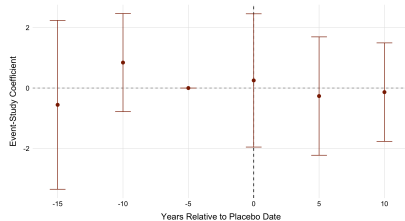


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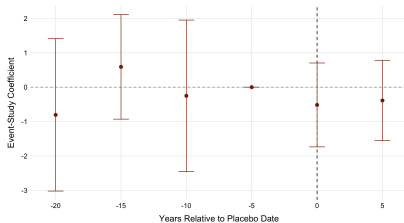
Placebo event studies using alternative pre-emancipation treatment dates



Placebo date: 1845

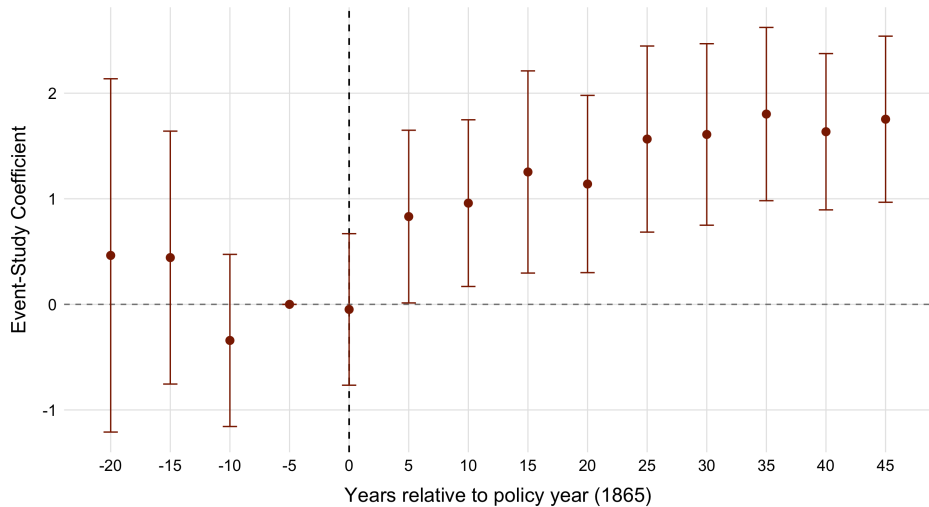


Placebo date: 1850



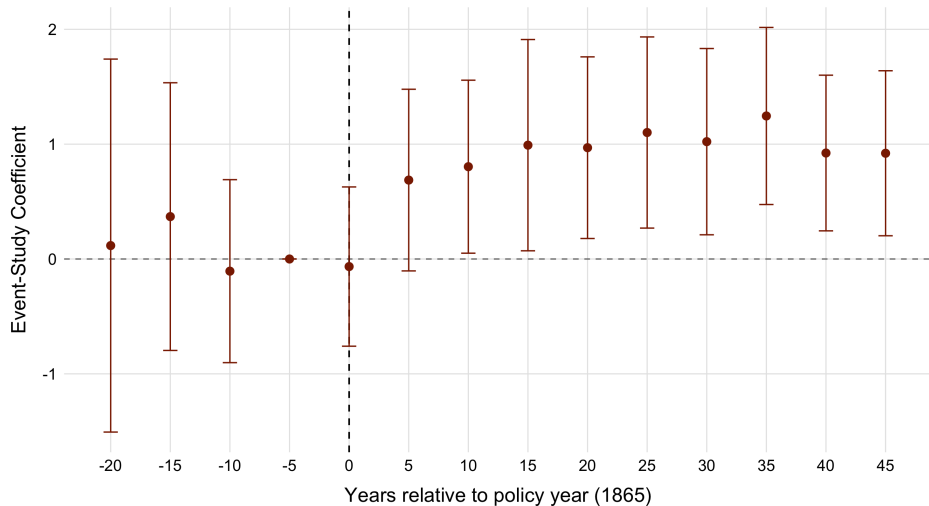
Placebo date: 1855

Alternative Control group: Hay/Grain harvesting



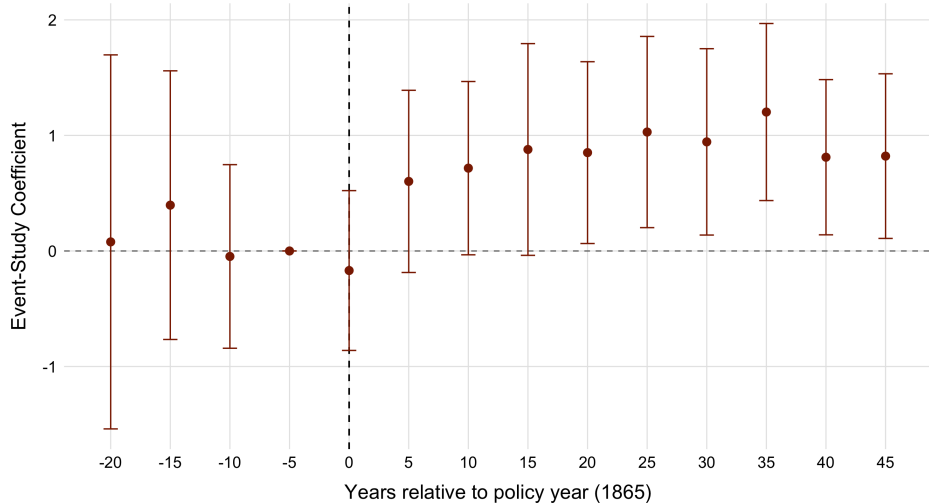
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Alternative Control group: All Agriculture, Husbandry, Food

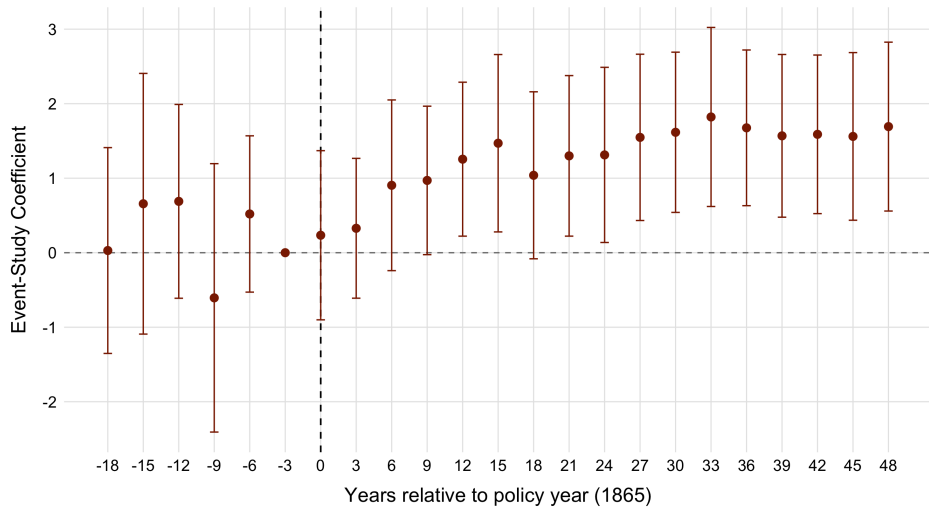


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Alternative Control group: All Agriculture, Husbandry, Food, Earth Working & wells

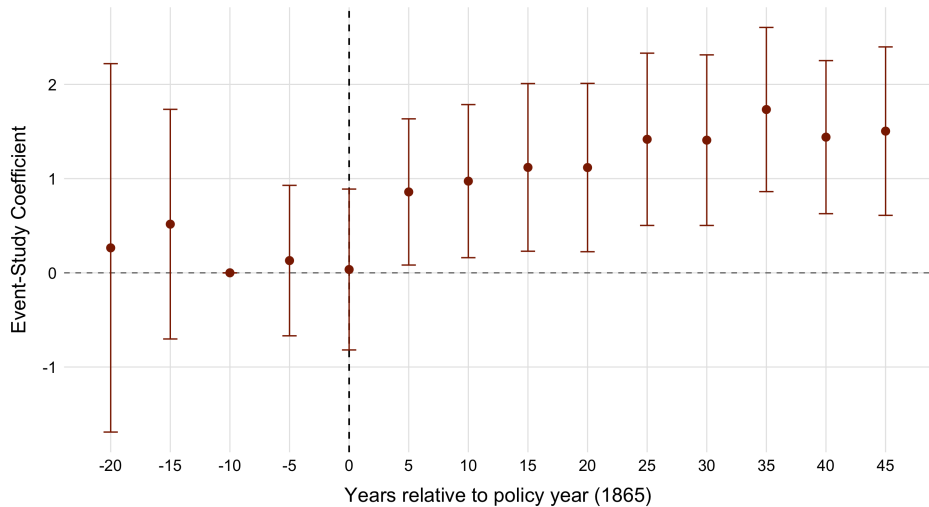


Robustness: 3 years Bin



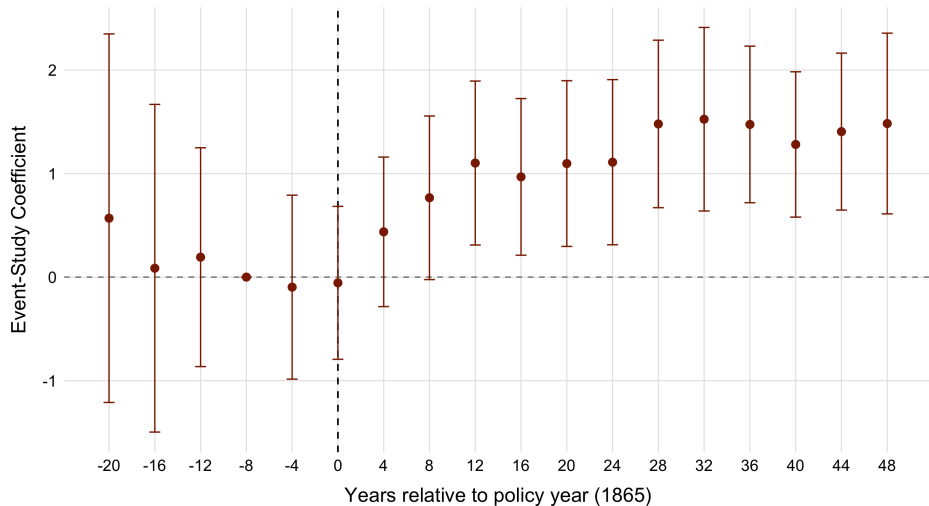
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Robustness: Reference Period=-2 (1855-1859)



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Robustness: 4 years Bin Reference Period=-2 (1857-1860)



Appendix: Productivity terms

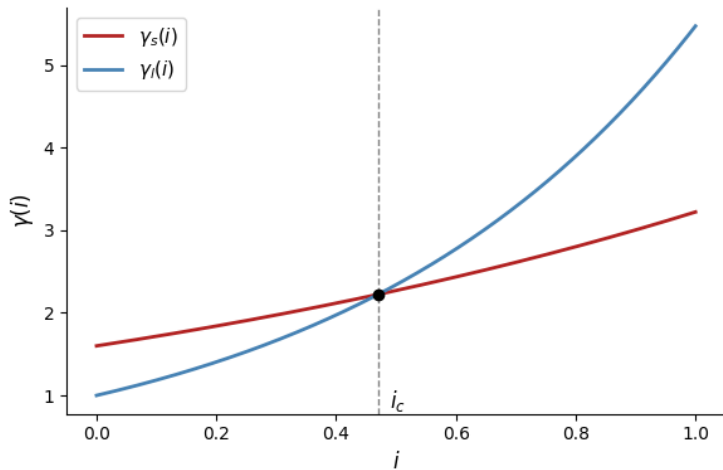


Figure 11

- USPTO Historical Patent Data
 - *Marco et al. (2019)*
- U.S. Census Data
 - *Haines (2004)*
- Census of Agriculture
 - *Haines, Fishback, & Rhode (2015)*
- Census of Manufactures
 - *Atack, Bateman, & Weiss (2004)*
 - *Hornbeck et al. (2025)*

Households maximize

$$U(t) = \int_t^{\infty} \exp(-\rho t) \ln C(t) dt, \quad (2)$$

subject to:

$$\dot{\mathcal{A}}_t + C_t \leq r_t \mathcal{A}_t + w_{Lt} L + w_{Ht} H,$$

and the usual non-Ponzi condition $\int_0^{\infty} \exp(-r_t t) \mathcal{A}_t dt \geq 0$, where $\mathcal{A}_t = \int_0^{I_t} V_t(i) di$.

This maximization problem of the representative household gives the usual Euler equation:

$$\frac{\dot{C}}{C} = r_t - \rho. \quad (3)$$

Firm Optimization

From final good sector:

$$p_t(i)y_t(i) = Y_t . \quad (4)$$

$$\text{Input choice } i : \min \left\{ \overbrace{\frac{\psi w_{lt}}{q_0(i)}}^{\text{cost of capital}}, \quad \overbrace{\frac{w_{st}}{\gamma_s(i)}}^{\text{enslaved rental rate}}, \quad \overbrace{\frac{w_{lt}}{\gamma_l(i)}}^{\text{free rental rate}} \right\}$$

$$\frac{w_{st}}{\gamma_s(i)} \leq \frac{w_{lt}}{\gamma_l(i)} \quad \text{for } i < \bar{l}, \quad \text{and} \quad \frac{w_{st}}{\gamma_s(\bar{l})} = \frac{w_{lt}}{\gamma_l(\bar{l})}. \quad (5)$$

Capital good k is produced by using ψl_t of the free labor.

$$\frac{\psi w_{lt}}{q_0(i)} < \frac{w_{st}}{\gamma_s(i)} \quad \text{for } i < l^*, \quad \text{and} \quad \frac{\psi w_{lt}}{q_0} = \frac{w_{st}}{\gamma_s(l^*)}. \quad (6)$$

Equilibrium prices: Capital intensive tasks

- capital good producer with the highest quality becomes monopolist
- Each innovation improves the quality of capital input in task i by $1 + \lambda$, such that the frontier quality becomes $q_{t+\Delta t}(i) = (1 + \lambda)q_t$.
- There are competitive fringe firms that have access to a lower productive vintage that can imitate the latest technology by the rate $\phi \in [\frac{1}{1+\lambda}, 1)$

$$\text{Limit pricing: } p(i) = \frac{\psi w_{lt}}{\phi q_t(i)}. \quad (7)$$

$$\text{capital demand: } k(i) = \frac{\phi Y_t}{\psi w_{lt}} \quad (8)$$

$$\text{Machine-building labor demand } L_{kt} = \int_0^{l_t} \psi k_t(i) di = \phi l_t Y_t / w_{lt}. \quad (9)$$

Equilibrium prices: Labor intensive tasks

Labor-intensive non-mechanized tasks are produced under the competitive markets:

$$p_t(i) = \begin{cases} \frac{w_{Lt}}{A_L c(i) \gamma(i)}, & \text{for all } l^* < i \leq \bar{l}; \\ \frac{w_{Ht}}{A_H \gamma(i)}, & \text{for all } i > \bar{l}. \end{cases} \quad (10)$$

$$l(i) = \frac{Y_t}{w_{st}} \quad \text{and} \quad h(i) = \frac{Y_t}{w_{lt}}. \quad (11)$$

From market clearing conditions $\int_0^l l_t(i) di = L$, $\int_0^1 h_t(i) di + L_{kt} = L$, $L_{kt} := \int_0^{l^*} \psi k_t(i) di$ and under the law of one price, rental rate of labors are given by:

$$w_{Lt} = [\bar{l} - l^*] \frac{Y_t}{L} \quad \text{and} \quad w_{lt} = [1 - \bar{l} + \phi l^*] \frac{Y_t}{H}. \quad (12)$$

Prices and Market Clearing

Cobb–Douglas aggregation across tasks implies equal expenditure:

$$p_t(i)y_t(i) = Y_t.$$

Hence, conditional on the input used,

$$k_t(i) = \frac{\phi Y_t}{\psi w_{H,t}}, \quad l_t(i) = \frac{Y_t}{w_{L,t}}, \quad h_t(i) = \frac{Y_t}{w_{H,t}},$$

From market clearing conditions

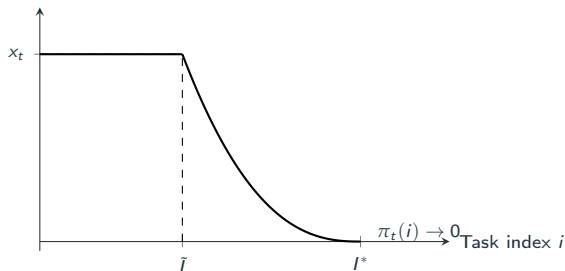
$\int_0^I l_t(i) di = L$, $\int_0^1 h_t(i) di + H_{kt} = H$, $H_{kt} := \int_0^{I^*} \psi k_t(i) di$ and under the law of one price, rental rate of labors are given by:

$$w_{L,t} = [\bar{I} - I^*] \frac{Y_t}{L}, \quad w_{H,t} = [1 - \bar{I} + \phi I^*] \frac{Y_t}{H}.$$

Limit pricing: $p_t(l_t) = \min \left\{ \frac{w_{st}}{\gamma_s(l_t)}, \frac{\psi w_{lt}}{\phi q_0(i)} \right\}$ for $l_t \neq l_{t-\Delta t}$. (13)

$$\pi_t(i) = \begin{cases} (1 - \phi) Y_t, & \text{for all } i \leq \tilde{l}; \\ \left(1 - \frac{\psi \gamma_s(i) w_{lt}}{q_0(i) w_{st}}\right) Y_t, & \text{for all } \tilde{l} < i \leq l^*. \end{cases} \quad (14)$$

Automation possibility



Now by integration and summing over all tasks, we obtain the final output,

$$\begin{aligned}\ln Y_t &= \underbrace{\int_0^{l_t} (\ln q_t(i) + \ln k(i)) di}_{\text{Machines}} + \underbrace{\int_{l_t}^{\bar{l}_t} (\ln \gamma_s(i) + \ln s(i)) di}_{\text{Enslaved labor}} + \underbrace{\int_{\bar{l}_t}^1 (\ln \gamma_l(i) + \ln l(i)) di}_{\text{Free labor}} \\ &= \int_0^{l_t} \left(\ln q_t(i) + \ln \frac{\phi Y_t}{\psi w_{lt}} \right) di + \int_{l_t}^{\bar{l}_t} \left(\ln \gamma_s(i) + \ln \frac{S}{\bar{l}_t - l_t} \right) di + \int_{\bar{l}_t}^1 \left(\ln \gamma_l(i) + \ln \frac{L}{1 - \bar{l}_t} \right) di\end{aligned}$$

Now by integration and summing over all tasks, we obtain the final output,

$$\begin{aligned}
 \ln Y_t &= \underbrace{\int_0^{l_t} (\ln q_t(i) + \ln k(i)) di}_{\text{Machines}} + \underbrace{\int_{l_t}^{\bar{l}_t} (\ln \gamma_s(i) + \ln s(i)) di}_{\text{Enslaved labor}} + \underbrace{\int_{\bar{l}_t}^1 (\ln \gamma_l(i) + \ln l(i)) di}_{\text{Free labor}} \\
 &= \int_0^{l_t} \left(\ln q_t(i) + \ln \frac{\phi Y_t}{\psi w_{lt}} \right) di + \int_{l_t}^{\bar{l}_t} \left(\ln \gamma_s(i) + \ln \frac{S}{\bar{l}_t - l_t} \right) di + \int_{\bar{l}_t}^1 \left(\ln \gamma_l(i) + \ln \frac{L}{1 - \bar{l}_t} \right) di \\
 Y_t &= Q_t \left(\frac{S}{\bar{l}_t - l_t} \right)^{\bar{l}_t - l_t} \left(\frac{L}{1 - \bar{l}_t} \right)^{1 - (\bar{l}_t - l_t)}, \tag{15}
 \end{aligned}$$

where $Q_t = \exp \left(\int_0^{l_t} \ln q_t(i) di + \int_{l_t}^{\bar{l}_t} \ln \gamma_s(i) di + \int_{\bar{l}_t}^1 \ln \gamma_l(i) di \right) \left(\frac{\phi}{\psi} \right)^{\frac{l_t}{1 - \bar{l}_t}}$ is the TFP term.

$$\frac{w_{st}}{\gamma_s(\bar{I})} = \frac{w_{lt}}{\gamma_l(\bar{I})}, \text{ and } \frac{\psi w_{lt}}{q_0(I^*)} = \frac{w_{st}}{\gamma_s(I^*)}.$$

$$\text{Free equilibrium cutoff } q_0(I^{**}) = \psi \gamma_l(I^{**}) \quad (16)$$

$$\text{Slave economy machine cutoff } \frac{\psi \gamma_s(I^*)}{q_0(I^*)} = \frac{\gamma_s(\bar{I})}{\gamma_l(\bar{I})} \quad (17)$$

$$\text{Slave economy labor cutoff } \frac{(\bar{I} - I^*)L}{(1 - \bar{I} + \phi I^*)S} = \frac{\gamma_s(\bar{I})}{\gamma_l(\bar{I})} \quad (18)$$

$$\text{Condition for the interior equilibrium } \frac{\psi \gamma_s(0)}{\gamma_s(\bar{I}_0)/\gamma_l(\bar{I}_0)} < q_0(0) < \frac{\psi \gamma_s(0)}{\gamma_s(1)/\gamma_l(1)}. \quad (19)$$

Appendix: Wage Premium

Table 1. Cost-premium for hiring free laborers relative to slave laborers (average 1824–1860), by length of contract and region.

Contract length	Region	Slave hire data source	Mean premium	Years of data	Min premium	Max premium
Yearly	Lower South	<i>Evans</i>	+73 %	20	−6 %	+153 %
		<i>Fogel & Engerman</i>	+219 %	32	+59 %	+402 %
	Upper South	<i>Evans</i>	+69 %	11	+22 %	+143 %
		<i>Fogel & Engerman</i>	+140 %	15	+79 %	+202 %
Monthly	Lower South	<i>Evans</i>	+35 %	26	−31 %	+114 %
		<i>Fogel & Engerman</i>	+437 %	22	+88 %	+1,045 %
		<i>Margo</i>	+16 %	16	−38 %	+146 %
	Upper South	<i>Evans</i>	+52 %	12	+2 %	+105 %
		<i>Fogel & Engerman</i>	+233 %	10	−21 %	+862 %
		<i>Margo</i>	+44 %	16	−30 %	+320 %
Daily	Lower South	<i>Evans</i>	+44 %	13	0 %	+78 %
		<i>Margo</i>	+44 %	16	−30 %	+320 %
	Upper South	<i>Evans</i>	+47 %	11	−30 %	+123 %

Sources: see [Figures 1–2](#). Note: All figures are compared to daily or monthly wages paid to free laborers from the same region from Margo’s dataset. Yearly slave hire prices are here compared to 12 monthly wages to free laborers. Pairs of data-series with fewer than 10 years of comparable data are excluded from the table.

Figure 13: Ronnback (2021)

Panel structure

- 5-year bins, 1845–1910 (14 periods)
- Unit of observation: subclass $\times$ period cell
- $N = 20,202$ cells

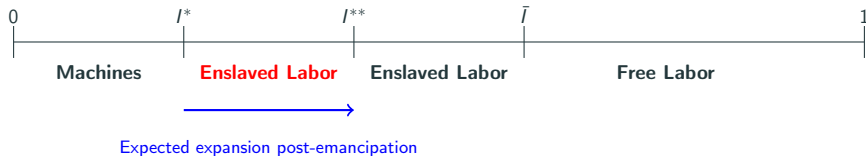
Subclasses (clusters)

- Treated: 61
- Control: 1,382
- Total: 1,443

	Treated	Control
Subclasses	61	1,382
Patents (total)	968	40,554
Total	41,522 patents	

Poisson QMLE with subclass and period FE;
SE clustered at subclass level.

Illustrative Idea



- **Innovation:** Emancipation triggers innovation in tasks $l^{**} - l^*$

◀ exogeneity

adoption

Unlike Patent data 2 problems:

1. Machine Availability

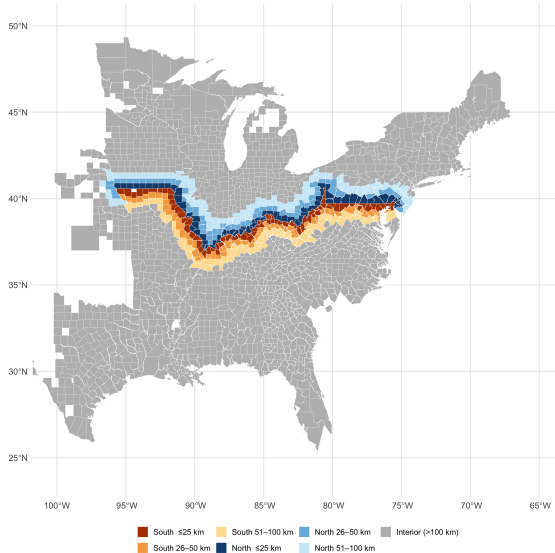
- Commercial cotton or tobacco harvester did not arrive until 1940s

2. Civil war affect

- Physical Capital Destruction (Feigenbaum et al., 2022)
- Capital Market Collapse (Bodenhorn and Rockoff, 1992) $\implies$ cost of capital $\uparrow$

predictions

Empirical Analysis at the North-South Border



Empirical Analysis at the North-South Border

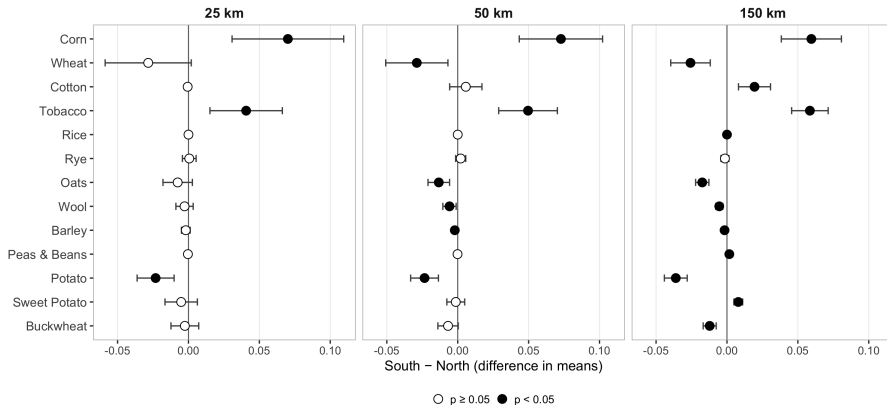


Figure 15: Mason-Dixon Border Line

Empirical Strategy: DDD Event Study

- **Data:** US Census of Agriculture & Manufactures
- **25 km buffer:** 77 Southern, 71 Northern counties **100 km buffer:** 222 Southern, 234 Northern counties

Triple Difference Event Study:

$$y_{cst} = \alpha_{cs} + \lambda_t + \sum_{k \neq -5} \beta_k \cdot 1[k = t] \cdot \text{South}_c \cdot \text{Farm}_s + \varepsilon_{cst}$$

y_{cst} Log capital share in county c , sector s , decade t

α_{cs} County $\times$ sector FE λ_t Decade FE

$$\hat{\beta}_k = \left[(\bar{y}_{S,F,k} - \bar{y}_{N,F,k}) - (\bar{y}_{S,M,k} - \bar{y}_{N,M,k}) \right] - \left[(\bar{y}_{S,F,1860} - \bar{y}_{N,F,1860}) - (\bar{y}_{S,M,1860} - \bar{y}_{N,M,1860}) \right]$$

Event Study: Dynamic Triple Difference Estimates

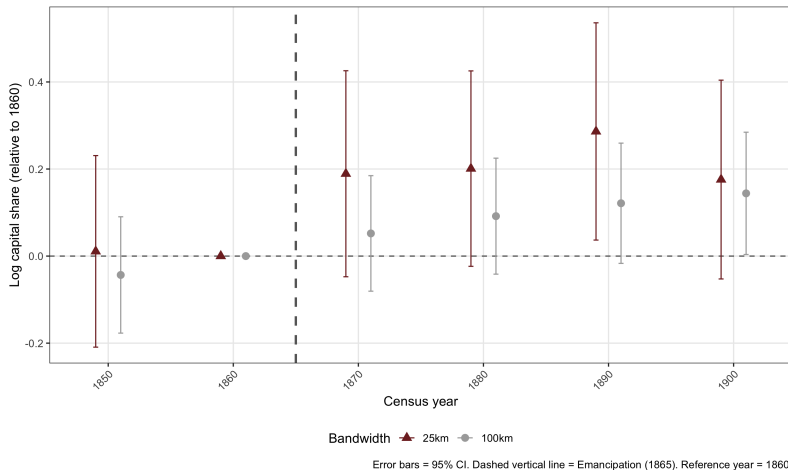


Figure 16: Event Study: Dynamic Triple Difference Estimates

Event Study: Dynamic Triple Difference Estimates with HM

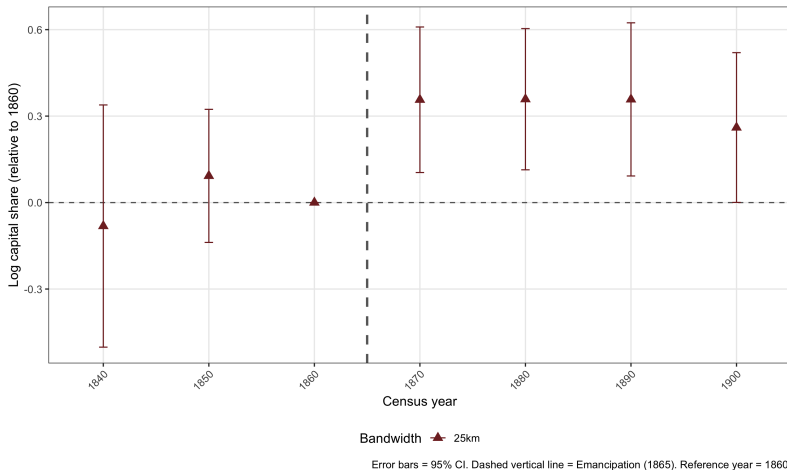


Figure 17: Event Study: Dynamic Triple Difference Estimates with HM

Empirical Strategy I: Triple Difference

Dependent variable: Log capital share in county c , sector s , year t :

$$y_{cst} = \begin{cases} \log\left(\frac{\text{Equipment value}}{\text{Ag. output}}\right)_{ct} & \text{if } s = \text{Farm} \\ \log\left(\frac{\text{Mfg. capital}}{\text{Mfg. output}}\right)_{ct} & \text{if } s = \text{Mfg.} \end{cases}$$

Estimating equation:

$$y_{cst} = \alpha_{cs} + \lambda_t + \delta (\text{Post}_t \times \text{South}_c \times \text{Farm}_s) + \varepsilon_{cst}$$

α_{cs} County $\times$ sector Fixed Effect

λ_t Year Fixed Effect

Post_t = 1 if year > 1861

South_c = 1 if county in former slave state

Farm_s = 1 if farm machinery sector, = 0 if manufacturing

Sample: Border counties (25km, 50km, 100km bands). Standard errors clustered at county level.

Table 6: Capital Reallocation: Triple Difference Estimates

		log_share		
	Full sample	25km	50km	100km
	(1)	(2)	(3)	(4)
Post × Treat × Farm machinery	-0.2575*** (0.0309)	0.2090** (0.0861)	0.1267** (0.0638)	0.1240** (0.0505)
Observations	20,992	1,609	3,000	4,983
R ²	0.83760	0.86198	0.85872	0.86305
county_sector fixed effects	✓	✓	✓	✓
YEAR fixed effects	✓	✓	✓	✓

Clustered standard errors at the county level in parentheses. All specifications include county-sector and year

Table 7: DD Estimates: Farm Capital Share

	log_share			
	Full	25km	50km	100km
	(1)	(2)	(3)	(4)
Post $\times$ Treat	-0.3974*** (0.0187)	0.0430 (0.0313)	0.0123 (0.0245)	0.0101 (0.0228)
<i>Panel A: No controls</i>				
Controls	No	No	No	No
Observations	12,547	884	1,630	2,689
R ²	0.66368	0.79013	0.77602	0.78132
county_id fixed effects	✓	✓	✓	✓
YEAR fixed effects	✓	✓	✓	✓

Results

	log_share			
	Full	25km	50km	100km
	(1)	(2)	(3)	(4)
Post $\times$ Treat	-0.4052*** (0.0196)	0.0408 (0.0314)	0.0121 (0.0243)	0.0048 (0.0225)
<i>Panel B: AVGIMP control</i>				
Controls	Yes	Yes	Yes	Yes
Observations	12,513	883	1,625	2,678
R ²	0.66745	0.79024	0.77625	0.78222
county_id fixed effects	✓	✓	✓	✓
YEAR fixed effects	✓	✓	✓	✓

Horses and Mules (HM) vs Farm Equipment

◀ Back

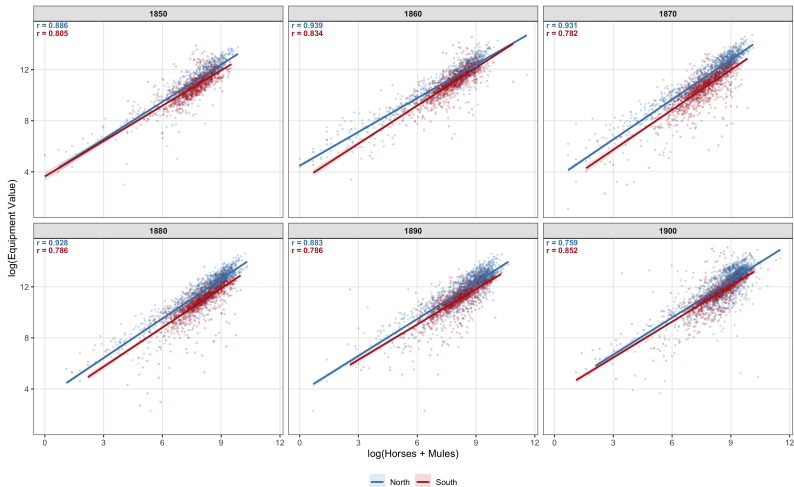


Figure 18: Correlation between $\log(\text{Farm equipment value})$ and $\log(\text{Horses} + \text{Mules})$.

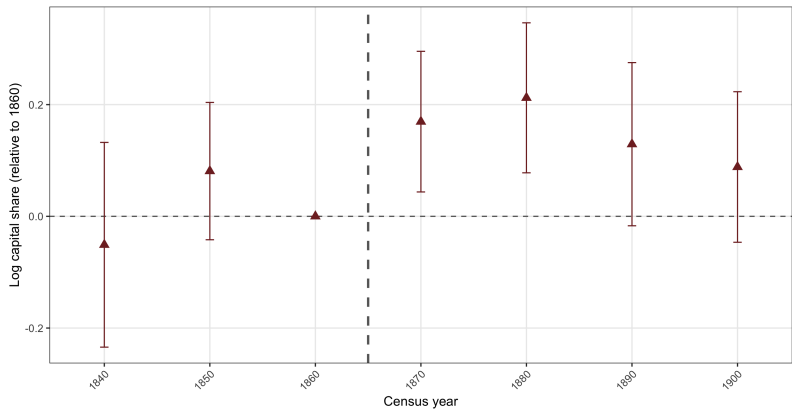
Table 8: DiD Estimates: Horses and Mules Share (1840–1900)

	log_hm_share			
	Full	25km	50km	100km
	(1)	(2)	(3)	(4)
Post × Treat	-0.1394*** (0.0192)	0.1354*** (0.0462)	0.0976*** (0.0333)	0.0360 (0.0261)
Observations	13,801	1,000	1,833	3,033
R ²	0.74294	0.77118	0.78233	0.78547
county_id fixed effects	✓	✓	✓	✓
YEAR fixed effects	✓	✓	✓	✓

Clustered standard errors at the county level in parentheses. All specifications include county and year fixed effects. Outcome is $\log(\text{Horses} + \text{Mules} / \text{Agricultural Output})$. Sample extends to 1840. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Diff-in-Diff Horses and Mules

◀ Back



Bandwidth ▲ 25km

Error bars = 95% CI. Dashed vertical line = Emancipation (1865). Reference year = 1860.

Figure 19: Mason-Dixon Border Line

Event Study: Dynamic Triple Difference Estimates with HM

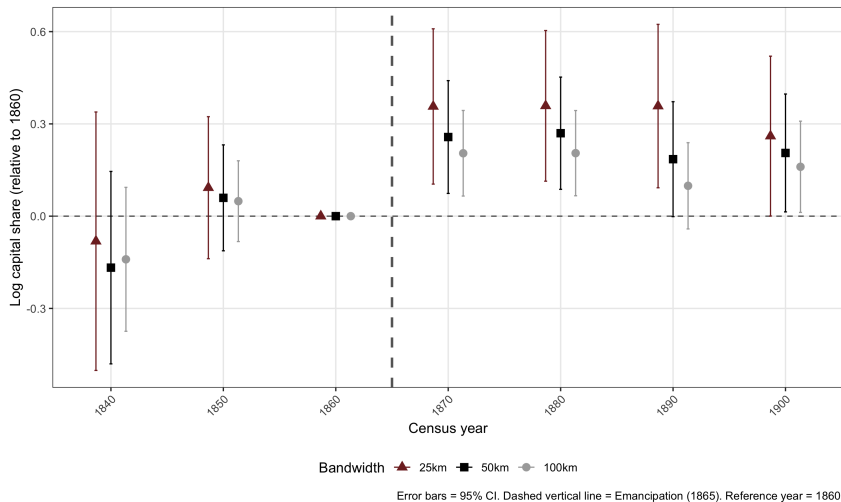


Figure 20: Event Study: Dynamic Triple Difference Estimates with HM for 25km, 50km, and 100km bands

Table 9: Capital Reallocation: Triple Difference Estimates: Horses and Mules (1840–1900)

	log_share			
	Full sample	25km	50km	100km
	(1)	(2)	(3)	(4)
Post × Treat × Farm (HM)	-0.0008 (0.0317)	0.3254*** (0.0921)	0.2576*** (0.0677)	0.1898*** (0.0511)
Observations	23,373	1,840	3,402	5,659
R ²	0.97220	0.97688	0.97662	0.97699
county_sector fixed effects	✓	✓	✓	✓
YEAR fixed effects	✓	✓	✓	✓

Clustered standard errors at the county level in parentheses. All specifications include county-sector and year fixed effects. Farm sector uses horses and mules per agricultural output. Sample extends to 1840. *p<0.10, **p<0.05, ***p<0.01.

Two representative households

Both households consume the national final good (the numeraire). Household $f \in \{N, S\}$ solves

$$\max U^f = \int_0^\infty e^{-\rho t} \ln C_t^f dt \quad (1)$$

subject to the flow budget constraints

$$\dot{A}_t^N + C_t^N \leq r_t^N A_t^N + w_{ft}^N L^N, \quad \dot{A}_t^S + C_t^S \leq r_t^S A_t^S + w_{ft}^S L^S + w_{st} S, \quad (2-3)$$

the no-Ponzi condition $\lim_{t \rightarrow \infty} A_t^f \exp(-\int_0^t r_u^f du) \geq 0$, where

$$A_t^f = \int_0^{I^{f*}} V_t^f(i) di. \quad (4)$$

Endowments are supplied inelastically; the Southern household owns S , rents it at w_{st} , and keeps the income. Each problem yields

$$\frac{\dot{C}_t^f}{C_t^f} = r_t^f - \rho, \quad f \in \{N, S\}. \quad (5)$$

Prices and quantities within a region

The log task aggregator $\ln Y_t^f = \int_0^1 \ln y_t^f(i) di$ implies *equal expenditure shares across tasks*:

$$p_t^f(i) y_t^f(i) = E_t^f, \quad E_t^f \equiv P_t^f Y_t^f. \quad (14)$$

- Every within-region equation of the one-region model survives, with national output Y_t replaced by **regional revenue** E_t^f .
- In the South only, a labour–labour cutoff $\bar{l}_t$ separates enslaved from free tasks:

$$\frac{w_{st}}{w_{ft}^S} = R(\bar{l}_t). \quad (15)$$

No analogue in the North.

Marginal-cost pricing of labour tasks and limit pricing against a fringe of quality $\phi q_t^f(i)$ give

$$s_t(i) = \frac{E_t^S}{w_{st}}, \quad \ell_t^f(i) = \frac{E_t^f}{w_{ft}^f}, \quad p_t^f(i) = \frac{\psi w_{ft}^f}{\phi q_t^f(i)}, \quad k_t^f(i) = \frac{\phi E_t^f}{\psi w_{ft}^f}. \quad (16)$$

Factor market clearing and regional wages

Imposing $\int_0^1 s_t(i) di = S$ and $\int_0^1 \ell_t^f(i) di + L_{kt}^f = L^f$ on (??):

$$w_{ft}^N = [1 - (1 - \phi)I_t^N] \frac{E_t^N}{L^N},$$
$$w_{st} = (\bar{I}_t - I_t^S) \frac{E_t^S}{S}, \quad w_{ft}^S = (1 - \bar{I}_t + \phi I_t^S) \frac{E_t^S}{L^S}. \quad (17)$$

Regional output and regional growth

Aggregating the tasks, with the convention $\bar{l}_t^N \equiv l_t^N$:

$$Y_t^f = Q_t^f \left(\frac{L^f}{1 - \bar{l}_t^f + \phi l_t^f} \right)^{1 - (\bar{l}_t^f - l_t^f)} \left(\frac{S^f}{\bar{l}_t^f - l_t^f} \right)^{\bar{l}_t^f - l_t^f}, \quad (18)$$

with $S^N = 0$, $S^S = S$, and the regional quality index

$$Q_t^f = \exp \left(\int_0^{l_t^f} \ln q_t^f(i) di + \int_{l_t^f}^{\bar{l}_t^f} \ln \gamma_s(i) di + \int_{\bar{l}_t^f}^1 \ln \gamma_\ell(i) di \right) \left(\frac{\phi}{\psi} \right)^{l_t^f}.$$

- North collapses to $Y_t^N = Q_t^N L^N / [1 - (1 - \phi) l_t^N]$.
- Endowments L^f and the cutoffs are constant at the asymptotic allocation, so **all growth is quality growth**:

$$g_t^f \equiv \frac{\dot{Y}_t^f}{Y_t^f} = \frac{\dot{Q}_t^f}{Q_t^f}. \quad (19)$$

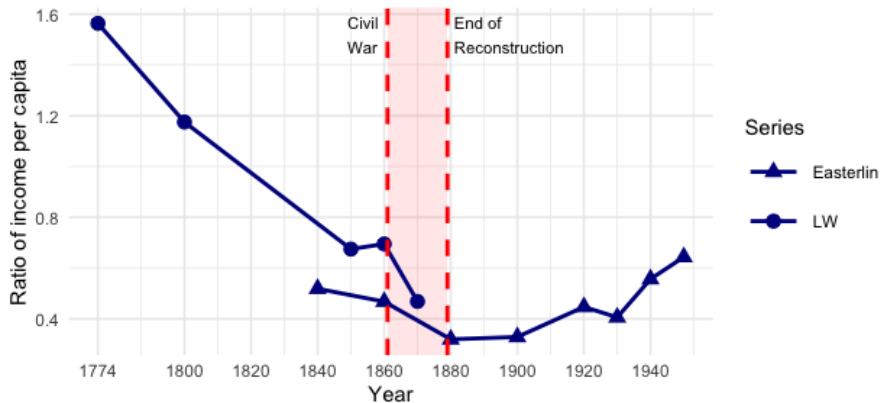
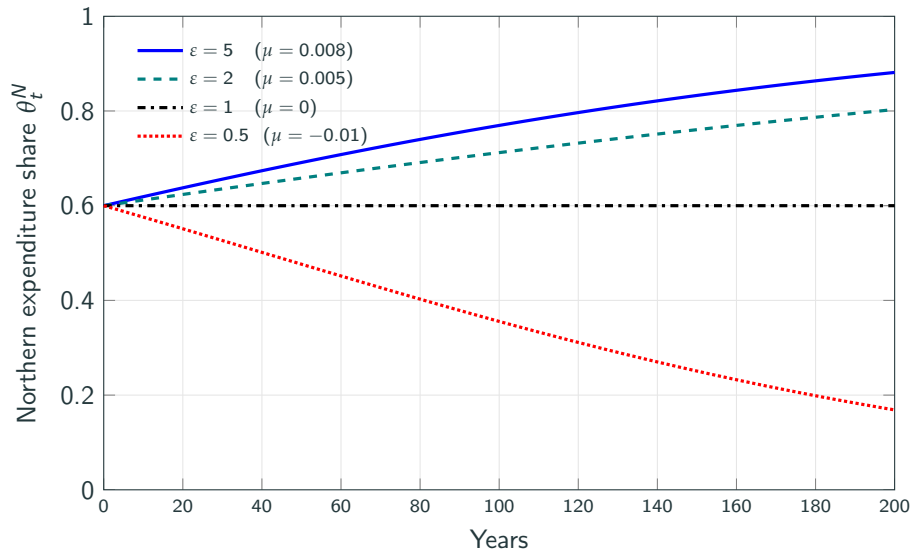


Figure 21: Ratio of income per capita South/North between original thirteen states

The Share transition



HJB:

$$r_t V_t(i) - \dot{V}_t(i) = \pi_t(i) - \tau_t(i) V_t(i)$$

Entrant Problem: $\max_z \{z_t(i) V_t(i) - \kappa_I z_t(i)^\alpha Y_t\} \Rightarrow z_t(i) = \left[\frac{V_t(i)}{\kappa_I \alpha Y_t} \right]^{\frac{1}{\alpha-1}}$

Mechanization occurs at the marginal task:

$$i_t = z_t(l_t) = \left[\frac{V_t(l_t)}{\kappa_I \alpha Y_t} \right]^{\frac{1}{\alpha-1}}, \quad l_t < l^{**}$$

$$\dot{i}_t = 0 \quad \text{at} \quad l_t = l^{**}.$$

As the value of mechanizing the marginal task falls, mechanization slows and stops at l^{**} .

Machine-Task productivity

$$q_0(i) = \bar{q} e^{-\beta q i}$$

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Enslaved Labor

$$\gamma_s(i) = C$$

Free Labor

$$\gamma_\ell(i) = \begin{cases} a, & i \leq i_f, \\ a + \beta_\ell (i - i_f), & i > i_f, \end{cases}$$

Mechanization and the Law of Motion

Gain from mechanizing the marginal task

$$\mu_H(I_t^S) = \min \left\{ 1 - \phi, 1 - \frac{\psi \gamma_I(I_t^S)}{q_0(I_t^S)} \right\}, \quad \mu_V \equiv 1 - \phi.$$

Horizontal R&D problem

$$R_t^H = \kappa_I z_t^\alpha E_t^S, \quad \dot{I}_t^S = z_t,$$

with free entry implying

$$\alpha \kappa_I z_t^{\alpha-1} = \frac{V_t^H}{E_t^S}, \quad \alpha \kappa_S (\tau_t^S)^{\alpha-1} = \frac{V_t^V}{E_t^S}.$$

Since

$$\frac{V_t^H}{V_t^V} = \frac{\mu_H(I_t^S)}{\mu_V},$$

the mechanization frontier evolves according to

$$\dot{I}_t^S = \left(\frac{\kappa_S}{\kappa_I} \right)^{\frac{1}{\alpha-1}} \tau_t^S \left[\frac{\mu_H(I_t^S)}{\mu_V} \right]^{\frac{1}{\alpha-1}}.$$

$$I_{1865}^S = I^* < I^{**}, \quad \dot{I}_t^S = \frac{\kappa_S}{\kappa_I} \tau_t^S m(I_t^S)$$

$$c_t = 1 - \kappa_S (\tau_t^S)^2 \left[I_t^S + \frac{\kappa_S}{\kappa_I} m(I_t^S)^2 \right]$$

$$\dot{\tau}_t^S = \frac{c_t \left[\tau_t^S (\rho + \tau_t^S) - \frac{1-\phi}{2\kappa_S} \right] - \kappa_S (\tau_t^S)^3 A'(I_t^S) \dot{I}_t^S}{1 + \kappa_S (\tau_t^S)^2 A(I_t^S)}, \quad A(I) = I + \frac{\kappa_S}{\kappa_I} m(I)^2$$

As I_t^S rises, profitability and the consumption share change, so τ_t^S adjusts along the transition. Eventually $I_t^S \rightarrow I^{**}$ and $\dot{I}_t^S \rightarrow 0$.

Table 10: Benchmark and Counterfactual Outcomes in 1900

Scenario	S/N GDP per worker	Δ vs. benchmark (percent)	South mech. I_S	Nominal S/N	Nati
<i>A. Data and estimated benchmark</i>					
Historical data	0.307	–	–	0.210	
Estimated benchmark	0.299	+0.0	0.311	0.206	
<i>B. Timing of abolition</i>					
Abolition in 1810	0.395	+31.9	0.328	0.257	
Abolition in 1830	0.369	+23.2	0.328	0.244	
Abolition in 1850	0.327	+9.2	0.327	0.221	
Abolition in 1880	0.277	-7.5	0.249	0.194	
<i>C. Institutional and transition counterfactuals</i>					
No abolition / permanent slavery	0.398	+33.0	0.166	0.259	
Abolition, no labor-supply decline	0.270	-9.7	0.311	0.190	
Static-only abolition	0.255	-15.0	0.166	0.181	

Notes: The historical row reports the 1900 CES-converted South/North real GDP-per-worker ratio, the observed nominal South/North ratio, and the observed national capital share for the selected capital-share series.

- **Capital share, 1800:** disciplines the initial level of mechanization in the North and South.
- **Capital shares, 1860, 1890, and 1900:** discipline the evolution of mechanization and the speed of convergence over time.
- **North–South relative GDP-per-worker growth, 1800–1860:** disciplines the gap between the coercive and free-economy mechanization frontiers, l^* and l^{**} .
- **Enslaved/free hire-cost ratio, 1865:** $w_H/w_L = 1.69$ disciplines the pre-abolition labor-allocation cutoff $\bar{l}$.
- **Long-run Northern growth:** pins down τ^* conditional on l^{**} .

1. **Guess parameters:** $\Theta = (I^{**}, I^*/I^{**}, \log \xi, f_N, f_S, \bar{I}, \beta_c)$, where $I^* = (I^*/I^{**})I^{**}$, $I_N(1800) = f_N I^{**}$, and $I_S(1800) = f_S I^*$.
2. **Recover technology and the BGP:** use the stationary cutoff conditions at I^{**} and I^* to recover the implied technology objects, and set $\tau^* = g_N^{ss}/(I^{**}\Lambda)$.
3. **Solve the off-BGP stable arms:** combine the HJB and mechanization laws to solve $d\tau/dI = \dot{\tau}/\dot{I}$, obtaining $\tau^F(I)$ and $\tau^C(I)$, together with $\bar{I}(I)$ under coercion.
4. **Map into historical time and simulate:** use $\dot{I} = (1/\xi)\tau(I)m(I)$ to map the stable arms into calendar time. The North follows the free arm; the South follows the coercive arm until 1865, when I_S is continuous and τ_S jumps to the free arm.
5. **Compute the seven model moments:** s_K^{1800} , s_K^{1860} , s_K^{1890} , s_K^{1900} , $\Delta \ln(y_S/y_N) * 1800 - 1860$, $\ln(w_H/w_L) * 1865$, and $\ln(Q^C/Q^F)_{1865}$.
6. **Update parameters:**

Functional forms and dynamics

- **Task productivity and machine quality:**

$$\gamma(i) = e^{\beta i}, \quad q_0(i) = \bar{q} e^{-\beta_q i} / i,$$

with $a \equiv \beta + \beta_q$ governing the slope of the mechanization frontier.

- **Coercion wedge:**

$$c(i) = C e^{-\beta_c i},$$

where β_c governs how the coercion advantage varies across tasks.

- **Mechanization dynamics:**

$$\dot{l}_o = \frac{1}{\xi} \tau_o m_o(l_o),$$

so ξ governs the speed of horizontal mechanization.

- **Innovation dynamics:** τ_o is a jump variable determined jointly with l_o from the HJB system.
- **Solution method:** solve the free and coercive stable arms backward from (l^{**}, τ^*) and (l^*, τ^*) .
- **Abolition:** l_S is predetermined in 1865, while τ_S jumps from the coercive to the free stable arm.

Calibration: matched moments

The moment-discrepancy vector is

$$g(\Theta) = \begin{bmatrix} \textit{capitalshare}^{1800}(\Theta) - 0.1448 \\ \textit{capitalshare}^{1860}(\Theta) - 0.2511 \\ \textit{capitalshare}^{1890}(\Theta) - 0.2834 \\ \textit{capitalshare}^{1900}(\Theta) - 0.2956 \\ \Delta \ln(y_S / y_N)_{1800-1860}(\Theta) - (-0.3473) \\ \ln(w_H / w_L)_{1865}(\Theta) - \ln(1.69) \\ \ln(Q^C / Q^F)_{1865}(\Theta) - \ln(1.35) \end{bmatrix}.$$

The parameter vector is chosen to minimize

$$\hat{\Theta} = \arg \min_{\Theta} g(\Theta)' W g(\Theta)$$

Targeted Moments

- Capital share 1800,1860,1890,1900
- 1800-1860 North South gdp per worker ratio
- slave free labor hiring rate premium

Abolition (1865)

1. Immediate productivity loss on the reallocated range:

$$\begin{aligned}\Delta \ln Q_{1865}^S = & \int_{I^*}^{\bar{I}_{pre}} \ln [A_L c(i) \gamma(i)] di + \int_{\bar{I}_{pre}}^1 \ln [A_H \gamma(i)] di \\ & - \int_{I^{**}}^{\bar{I}_{post}} \ln [A_L \gamma(i)] di - \int_{\bar{I}_{post}}^1 \ln [A_H \gamma(i)] di = \ln(1.35).\end{aligned}$$

2. Labor supply decline in the south L_S^{pre} decreased 32% $\implies L_S^{post}/H_S^{post} \downarrow$
3. Convergence to the new steady state, $I^* \longrightarrow I^{**}$.

Numerical solution and estimation

1. Guess a candidate parameter vector Θ .
2. Given Θ , solve the free and coercive stable arms: $\tau^F(I)$, $\tau^C(I)$, and $\bar{I}(I)$ under coercion.
3. Use the mechanization law of motion $\dot{I} = \frac{1}{\xi} \tau(I) m(I)$ to map the transition into historical time.
4. Simulate the North and South from their estimated 1800 positions to 1860, 1865, 1890, and 1900.
5. The North follows the free stable arm; the South follows the coercive arm until 1865 and the free arm afterward.
6. At abolition, I is predetermined and does not jump, while τ jumps from the coercive to the free stable arm at the same I .
7. From the simulated paths, compute the model moments $m^{model}(\Theta)$.
8. Compare them with the empirical targets using $g(\Theta) = m^{model}(\Theta) - m^{data}$.
9. Update the parameters and minimize $\hat{\Theta} = \arg \min_{\Theta} g(\Theta)' g(\Theta)$.

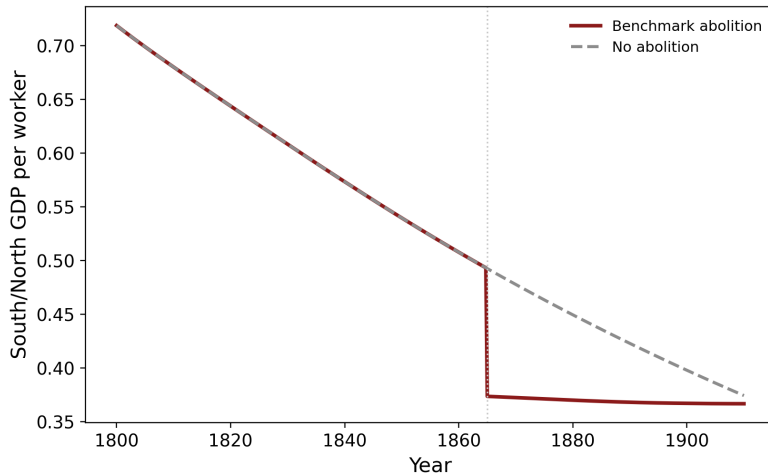


Figure 22: Model fit